Package ‘rENM.data’

June 21, 2026

Type Package

Title Data assembly tools for the rENM Framework

Version 0.1.0

Description Provides data acquisition and preparation tools for the rENM Framework, including extraction of eBird occurrence records and MERRA-2 environmental variables, spatial thinning, record limiting, extent determination, and temporal binning into five-year intervals.

Depends R (>= 4.1.0)

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Encoding UTF-8

Roxygen list(markdown = TRUE)

URL https://github.com/rENM-Framework/rENM.data

BugReports https://github.com/rENM-Framework/rENM.data/issues

Imports rENM.core,
future,
future.apply,
parallel,
sf,
stats,
terra,
tools,
utils

Config/roxygen2/version 8.0.0

Contents

find_occurrence_extent

Derive a spatial extent from occurrence data

Description

Pools valid coordinates from all per-bin occurrence CSV files in `_occs/`, computes a centered percentile bounding box, and writes the result to `_occs/extent.txt` for use in downstream rENM workflows.

Usage

```
find_occurrence_extent(alpha_code, bbox_pct = 99, project_dir = NULL)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code (e.g., "CASP").
`bbox_pct`	Numeric scalar. Central percentage of points to include in the bounding box. Must be in (0, 100]. Default is 99.
`project_dir`	Character. Path to the rENM project root. If `NULL` (default), resolved via `rENM_project_dir` (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

Details

Percentile bounding box

The bounding box is computed by selecting the central `bbox_pct` percent of valid points along each axis independently, discarding symmetric tails. For example, `bbox_pct = 95` uses the 2.5th and 97.5th quantiles; `bbox_pct = 100` uses the full coordinate range.

Coordinate validation

For each file, longitude and latitude columns are detected automatically from common aliases (`longitude`/`lon`/`x` and `latitude`/`lat`/`y`). Values are coerced to numeric; records outside the valid range (longitude -180 to 180, latitude -90 to 90) or exactly at (0, 0) are excluded. All surviving coordinates are pooled before quantile computation.

This function is typically called after `tidy_occurrences` has moved finalized occurrence files into `_occs/`.

Value

Invisibly returns a list with elements `alpha_code`, `bbox` (list with `xmin`, `xmax`, `ymin`, `ymax`), `paths` (list with `extent` and `log` file paths), and `counts` (list with `files`, `rows_read`, `points_used`, `bbox_pct`).

See Also

`tidy_occurrences`, `find_range_extent`, `set_extent`, `get_merra_variables`

Examples

```
## Not run:
res <- find_occurrence_extent("CASP")
res$bbox

res95 <- find_occurrence_extent("CASP", bbox_pct = 95)
find_occurrence_extent("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

`find_range_extent` *Derive a spatial extent from a species USGS GAP range polygon*

Description

Reads a USGS GAP range shapefile for a species, computes a padded geographic bounding box in WGS84 coordinates, and writes the result to `_occs/extent.txt` within the species run directory.

Usage

```
find_range_extent(alpha_code, pad_pct = 2, project_dir = NULL)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code matching the `ALPHA.CODE` field in the project species table.
`pad_pct`	Numeric scalar. Symmetric percent padding applied to the bounding box. Positive values expand the extent and negative values contract it. Must be greater than -100. Default is `2`.
`project_dir`	Character scalar or `NULL`. Path to the rENM project root. If `NULL`, the project directory is resolved using `rENM_project_dir()`.

Details

Existing `extent.txt` files are automatically backed up prior to writing a new extent definition.

The species lookup table is expected at `data/_species.csv` under the project root directory.

The function identifies the appropriate GAP range identifier using the `GAP.RANGE` field associated with the supplied `alpha_code`. The corresponding shapefile is expected at `data/shapefiles/GAP_RANGE/GAP_RANGE.s` under the project root, where `GAP_RANGE` is the value of that field.

If the shapefile coordinate reference system is not longitude/latitude WGS84 (EPSG:4326), it is transformed before extent calculation. The resulting bounding box is symmetrically padded relative to its center using `pad_pct`. The result is written to `runs/ALPHA_CODE/_occs/extent.txt`.

Value

Invisibly returns a named list with numeric elements `ul_lon`, `ul_lat`, `lr_lon`, and `lr_lat` (upper-left and lower-right corners in decimal degrees).

See Also

find_occurrence_extent, set_extent, get_merra_variables

Examples

```
## Not run:
res <- find_range_extent("CASP")

find_range_extent(
  alpha_code = "CASP",
  pad_pct = 5
)

## End(Not run)
```

---

get_ebird_occurrences *Read and bin an eBird EBD occurrence file by species*

Description

Reads a species' eBird Basic Dataset (EBD) text file, filters records to valid coordinates, writes a cleaned CSV copy, and organizes records into 5-year temporal bins for use in rENM workflows.

Usage

```
get_ebird_occurrences(alpha_code, project_dir = NULL)
```

Arguments

alpha_code	Character scalar. Four-letter species alpha code (e.g., "CASP").
project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Details

Input location

The EBD text file is located using the EBD.RECORDS field returned by get_species_info:

PROJECT_DIR/data/ebird/EBD_DATASET/EBD_DATASET.txt

Coordinate filtering

Longitude and latitude columns are detected automatically using common aliases. Records are retained only if coordinates are numeric, within valid ranges (longitude -180 to 180, latitude -90 to 90), and not exactly (0, 0). Duplicate records are retained at this stage.

Temporal binning

Observation dates are parsed to years and assigned to 5-year bins spanning 1980 to 2024 (1980, 1985, ..., 2020). Records outside this range are dropped. One CSV file per bin is written to _occs/tmp/ with the naming pattern of-YEAR.csv, containing species, longitude, and latitude columns.

Outputs

A cleaned copy of the full EBD file is written to RUN_DIR/_occs/EBD_DATASET.csv. Per-bin files go to RUN_DIR/_occs/tmp/. A processing summary is appended to RUN_DIR/_log.txt.

Value

Invisibly returns a list with elements csv_copy (path to cleaned CSV), bin_dir (tmp directory), bin_files (character vector of per-bin file paths), log_file, n_read (rows read from EBD), n_valid (rows with valid coordinates), and total_written (rows written across all bins).

See Also

rENM_project_dir, get_species_info, remove_duplicate_occurrences, thin_occurrences, tidy_occurrences

Examples

```
## Not run:
occ <- get_ebird_occurrences("CASP")
occ <- get_ebird_occurrences("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

---

get_merra_variables — *Crop and export MERRA-2 predictor rasters for a species run*

Description

For each 5-year bin from 1980 to 2020, locates MERRA-2 raster files from the m2 and mc source directories, crops them to the species' spatial extent, and writes the results to the run's _vars/ directory.

Usage

```
get_merra_variables(
  alpha_code,
  m2_vars = NULL,
  mc_vars = NULL,
  file_type = c(".tif", ".asc"),
  project_dir = NULL
)
```

Arguments

alpha_code	Character scalar. Four-letter species code.
m2_vars	Character vector or NULL. Variable basenames to include from the m2 directory. NULL includes all.
mc_vars	Character vector or NULL. Variable basenames to include from the mc directory. NULL includes all.
file_type	Character. Output format(s): ".tif", ".asc", or both. Default is both.

project_dir — Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Details

Input locations

Rasters (.tif, .tiff, or .asc) are read from:

- PROJECT_DIR/data/merra/m2/YEAR/
- PROJECT_DIR/data/merra/mc/YEAR/

When multiple files share the same basename, GeoTIFF is preferred over ASC. Use m2_vars and mc_vars to restrict processing to specific variable names (matched by basename, case-insensitive).

Spatial extent

The bounding box is read from RUN_DIR/_occs/extent.txt, which must be created first by find_occurrence_extent, find_range_extent, or set_extent.

CRS handling

Geographic rasters (lon/lat CRS or EPSG:4326) with 0-360 longitude are rotated to -180-180 before cropping. Projected rasters are cropped by projecting the WGS84 bounding box polygon into the raster CRS. Rasters that produce no valid cells after cropping are skipped.

Output

Cropped rasters are written to RUN_DIR/_vars/YEAR/. For .asc output, GDAL sidecar files (.aux.xml, .xml, .prj) are removed. Missing CRS on output is assigned WGS84.

Value

Invisibly returns a data frame with one row per candidate raster and columns year, family, basename, in_file, tif_written, asc_written, and notes (reason for skipping, or empty string on success).

See Also

find_occurrence_extent, find_range_extent, set_extent

Examples

```
## Not run:
get_merra_variables("CASP")
get_merra_variables("CASP", m2_vars = c("bio1", "bio12"),
                    project_dir = "/projects/rENM")

## End(Not run)
```

`limit_record_count` *Downsample occurrence records per bin to a maximum count*

Description

Randomly samples up to `record_count` rows from each per-bin occurrence CSV in `_occs/tmp/`, overwriting each file in place. Files with fewer rows than `record_count` are left unchanged. Reproducible sampling requires setting a seed externally with `set.seed()` before calling this function.

Usage

```
limit_record_count(alpha_code, record_count = 250, project_dir = NULL)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code (e.g., `"CASP"`).
`record_count`	Integer scalar. Maximum records to retain per bin. Must be positive. Default is 250.
`project_dir`	Character. Path to the rENM project root. If `NULL` (default), resolved via `rENM_project_dir` (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

Value

Invisibly returns a list with elements `alpha_code`, `files_processed`, `counts_before` (named integer vector), `counts_after` (named integer vector), `output_dir`, and `log_file`.

See Also

`get_ebird_occurrences`, `remove_duplicate_occurrences`, `thin_occurrences`, `tidy_occurrences`

Other occurrence processing: `remove_duplicate_occurrences()`, `thin_occurrences()`

Examples

```
## Not run:
limit_record_count("CASP", record_count = 250)
limit_record_count("CASP", record_count = 500, project_dir = "/projects/rENM")

## End(Not run)
```

`remove_duplicate_occurrences`
Remove duplicate occurrence records from per-bin CSV files

Description

Reads each per-bin occurrence CSV in `_occs/tmp/`, removes rows with identical longitude and latitude values (retaining the first occurrence of each unique coordinate pair), and overwrites the file in place.

Usage

```
remove_duplicate_occurrences(alpha_code, project_dir = NULL)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code (e.g., `"CASP"`).
`project_dir`	Character. Path to the rENM project root. If `NULL` (default), resolved via `rENM_project_dir` (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

Value

Invisibly returns a list with elements `alpha_code`, `files_processed` (character vector of processed file paths), `duplicates_removed` (named integer vector of counts per file), `output_dir`, and `log_file`.

See Also

`get_ebird_occurrences`, `thin_occurrences`, `tidy_occurrences`

Other occurrence processing: `limit_record_count()`, `thin_occurrences()`

Examples

```
## Not run:
remove_duplicate_occurrences("CASP")
remove_duplicate_occurrences("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

set_extent *Set the spatial extent for a species run*

Description

Writes an extent.txt file for a species run from explicit bounding box coordinates. Any existing extent.txt is renamed to a backup before writing. Use find_occurrence_extent or find_range_extent to derive the extent automatically from data rather than supplying coordinates manually.

Usage

```
set_extent(alpha_code, ul = c(-128, 49), lr = c(-66.5, 24), project_dir = NULL)
```

Arguments

alpha_code	Character scalar. Four-letter species alpha code (e.g., "CASP").
ul	Numeric vector of length 2. Upper-left corner c(lon, lat). Default c(-128.0, 49.0).
lr	Numeric vector of length 2. Lower-right corner c(lon, lat). Default c(-66.5, 24.0).
project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Details

Coordinates are provided as upper-left and lower-right corners in (longitude, latitude) order. Non-canonical orientation (e.g., upper-left latitude below lower-right latitude) is reported as a warning but not corrected. Valid ranges are longitude -180 to 180 and latitude -90 to 90.

An existing extent.txt is backed up as extent.txt.original, or as a timestamped variant if a backup already exists.

Value

Invisibly returns a list with elements alpha_code, extent (list with ul and lr named numeric vectors), paths (list with extent and log file paths), and notes (character vector of orientation warnings, or NULL).

See Also

find_occurrence_extent, find_range_extent

Examples

```
## Not run:
set_extent("CASP")
set_extent("AMRO", ul = c(-130, 55), lr = c(-60, 20))
set_extent("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

set_up_run *Initialize an rENM run directory structure*

Description

Creates the directory layout required for a single rENM modeling run identified by a four-letter species alpha code. Missing directories are created; existing directories are preserved.

Usage

```
set_up_run(alpha_code, project_dir = NULL)
```

Arguments

alpha_code	Character scalar. Four-letter species alpha code (e.g., "CASP").
project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Details

The run directory is constructed as PROJECT_DIR/runs/ALPHA_CODE and populated with the following structure:

```
runs/<alpha_code>/
  _occs/                      occurrence input/output files
  _vars/                      cropped predictor rasters
  Trends/
    centroids/
    suitability/
    variables/
  Summaries/                  plots, tables, and reports
  TimeSeries/
    <year>/
      model/                  model outputs
      occs/                   processed occurrences
      vars/                   environmental predictors
```

Time series directories are created for 5-year bins spanning 1980 to 2020. A run log is created or appended at RUN_DIR/_log.txt.

Value

Invisibly returns a list with elements alpha_code, run_dir, created (character vector of newly created directory paths), existing (character vector of pre-existing paths), and log_file.

See Also

rENM_project_dir

Examples

```
## Not run:
run <- set_up_run("CASP")
run <- set_up_run("CASP", project_dir = "/projects/rENM")
run$run_dir
run$created

## End(Not run)
```

`thin_occurrences` *Apply spatial thinning to occurrence records (sequential)*

Description

Applies a greedy sequential thinning algorithm to per-bin occurrence CSV files in `_occs/tmp/`, enforcing a minimum separation distance between retained points. Files are overwritten in place.

Usage

```
thin_occurrences(alpha_code, thin_distance, project_dir = NULL)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code (e.g., `"CASP"`).
`thin_distance`	Numeric scalar. Minimum separation distance in kilometers between retained points. Must be positive.
`project_dir`	Character. Path to the rENM project root. If `NULL` (default), resolved via `rENM_project_dir` (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

Details

The thinning algorithm processes records in input order. The first point is always retained. Each subsequent point is retained only if its Haversine distance to every previously retained point is at least `thin_distance` kilometers. This is a greedy nearest-neighbor filter and does not guarantee an optimal solution, but is fast and deterministic for a given input order.

Use `thin_occurrences2` for parallel execution on large datasets.

Value

Invisibly returns a list with elements `alpha_code`, `files_processed`, `points_before` (named integer vector), `points_after` (named integer vector), `output_dir`, and `log_file`.

See Also

`thin_occurrences2`, `remove_duplicate_occurrences`, `tidy_occurrences`

Other occurrence processing: `limit_record_count()`, `remove_duplicate_occurrences()`

Examples

```
## Not run:
thin_occurrences("CASP", thin_distance = 10)
thin_occurrences("CASP", thin_distance = 10, project_dir = "/projects/rENM")

## End(Not run)
```

`thin_occurrences2` *Apply spatial thinning to occurrence records (parallel)*

Description

Applies spatial thinning and optional record capping to per-bin occurrence CSV files using parallel execution via the **future** and **future.apply** packages. Files are overwritten in place. This is the parallel counterpart to `thin_occurrences` and is suited to larger datasets where per-bin files are numerous or large.

Usage

```
thin_occurrences2(
  alpha_code,
  radius = 1,
  records = 250,
  workers = NULL,
  project_dir = NULL
)
```

Arguments

`alpha_code`	Character scalar. Four-letter species alpha code (e.g., `"CASP"`).
`radius`	Numeric scalar. Minimum separation distance in kilometers. Use `0` to disable spatial thinning. Default is 1.
`records`	Integer scalar. Maximum records to retain per file after thinning. Use `0` to keep all surviving records. Default is 250.
`workers`	Integer scalar or `NULL`. Number of parallel workers. If `NULL` (default), all available logical cores are used.
`project_dir`	Character. Path to the rENM project root. If `NULL` (default), resolved via `rENM_project_dir` (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

Details

Spatial thinning

If `radius` > `0`, a greedy Haversine nearest-neighbor filter is applied per file: the first point is always retained; each subsequent point is kept only if it is at least `radius` km from all previously retained points. Earth radius is assumed to be 6371 km. Set `radius = 0` to skip spatial thinning.

Record cap

If records > 0, the thinned dataset is further randomly downsampled to at most records rows per file. Set records = 0 to keep all rows surviving thinning.

Parallel execution

Files are processed independently on separate workers using a future::multisession plan. If workers is NULL, all available logical cores are used. Both **future** and **future.apply** must be installed; the function stops with an informative error if they are not.

Value

Invisibly returns a data frame with one row per processed file and columns file (file name), before (rows before thinning), kept_after_radius (rows after spatial filter, or NA if disabled), kept_after_records (rows after record cap, or NA if not applied), and after (rows written to disk).

See Also

thin_occurrences, remove_duplicate_occurrences, tidy_occurrences

Examples

```
## Not run:
out <- thin_occurrences2("CASP", radius = 10, records = 500)
out
thin_occurrences2("CASP", radius = 10, records = 500,
                  project_dir = "/projects/rENM")

## End(Not run)
```

---

tidy_occurrences — *Finalize occurrence files by moving them from staging to the main directory*

Description

Copies all CSV files from _occs/tmp/ into _occs/, overwrites any existing files with matching names, and removes the temporary directory. This is the final cleanup step in the occurrence preprocessing pipeline, signaling that the files are ready for downstream modeling.

Usage

```
tidy_occurrences(alpha_code, project_dir = NULL)
```

Arguments

alpha_code	Character scalar. Four-letter species alpha code (e.g., "CASP").
project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Value

Invisibly returns a list with elements alpha_code, files_moved (character vector of destination paths), destination_dir, and log_file.

See Also

get_ebird_occurrences, remove_duplicate_occurrences, thin_occurrences

Examples

```
## Not run:
tidy_occurrences("CASP")
tidy_occurrences("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

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